Midterm 1 Review

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Course Syllabus and reading assignments:

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a. Introduction
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a. Time response analysis and specifications
b. Stability analysis
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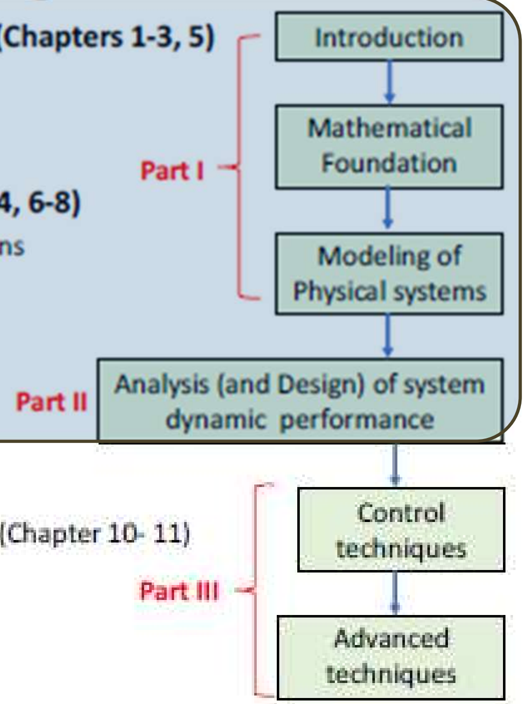
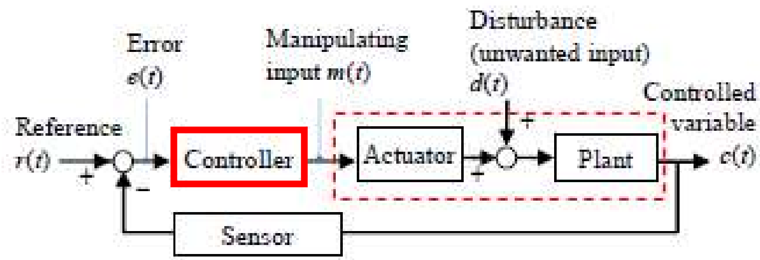




SISO Automatic (Closed-loop) Control System

Control objective: $c(t) \to r(t)$ regardless of $d(t)$.

Modeling in the Laplace Domain – Laplace Transform

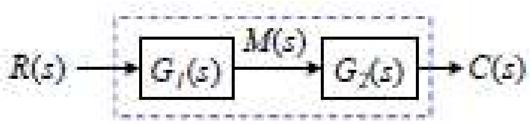
Laplace Transform (LT)

Complex variable $s = \underbrace{\sigma}_{\mathrm{Re}[s]} + \underbrace{j\omega}_{\mathrm{Im}[s]}$

Time Domain ⟷ **Laplace Domain**

$f(t)$ $\qquad \mathcal{L}[f(t)] = \int_0^\infty f(t)e^{-st}dt = F(s)$ $\qquad F(s)$

LT Operational symbol



$$\frac{C(s)}{R(s)} = G(s) = G_1 G_2 \cdots G_N(s)$$

Theorem

1. Linearity: $\mathcal{L}\{k_1 f_1(t) + k_2 f_2(t)\} = k_1 F_1(s) + k_2 F_2(s)$
2. Differentiation: $\mathcal{L}\left\{\frac{d^n f}{dt^n}\right\} = s^n F(s) - \sum_{k=1}^{n} s^{n-k}\left(\frac{d^{k-1}f}{dt^{k-1}}\right)_{t=0}$
3. Integration: $\mathcal{L}\left\{\int_0^t f(\tau)d\tau\right\} = \frac{F(s)}{s}$
4. Convolution: $\mathcal{L}\{f * g\} = F(s)G(s)$
5. **Final Value:** All poles must have negative real parts or be at the origin ➔ System must be **stable**
   $f(\infty) = \lim_{s \to 0} sF(s)$

	≑Capacitor-like	≑Inductor-like	≑Resistor-like
Elect. $e(t) \times i(t)$	$\frac{1}{Cs}$	Ls	R
Fluid $p(t) \times q(t)$	$\frac{\rho g}{A}$	$I_f s$	R_f

R, L, C in series:
$Z = Z_R + Z_L + Z_C$

R, L, C in parallel:
$\frac{1}{Z} = \frac{1}{Z_R} + \frac{1}{Z_L} + \frac{1}{Z_C}$

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Transfer Function - 1

$$G(s) = \frac{N(s)}{D(s)} = \frac{b_{m-1}s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

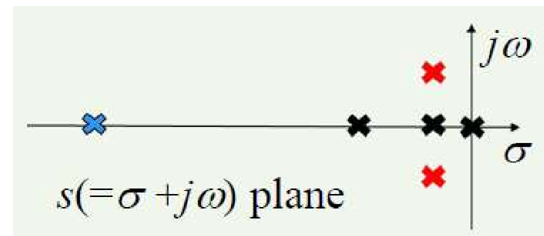
$$N(s) = 0 \Rightarrow \text{ Zeros}$$

$$D(s) = 0 \Rightarrow \text{ Poles}$$

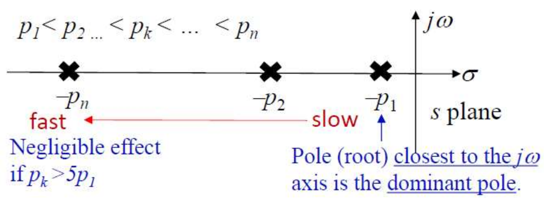
Characteristic Equation

Three types of poles

$$s = \underbrace{0, -1, -5,}_{\text{Real \& distinct}} \underbrace{-15, -15,}_{\text{Real \& Repetitive}} \underbrace{-1 \pm j}_{\text{Complex conjugate}}$$



- Stability and Speed of response based on the roots of D(s) = 0.

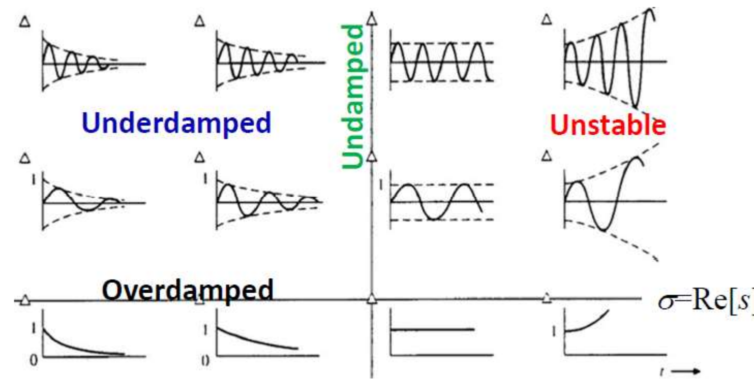


Transfer Function - 2

- First order system
  - $G(s) = \frac{C(s)}{R(s)} = \frac{1}{Ts+1}$, T= time constant, $c(t) = 0.632c_{ss}$
- Second order system
  - $G(s) = \frac{C(s)}{R(s)} = \frac{k}{ms^2+bs+k} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
  - $s_{1,2} = \frac{-b \pm \sqrt{b^2-4km}}{2m} = -\varsigma\omega_n \pm \omega_n\sqrt{\varsigma^2-1}$
  - $b^2 - 4km \begin{cases} >0 & \text{Real \& distinct} & (\varsigma > 1) \\ =0 & \text{Real, repetitive } (b_{\text{crit}} = 2\sqrt{km},\ \varsigma = 1) \\ <0 & \text{Complex conjugate} & (0 < \varsigma < 1) \end{cases}$

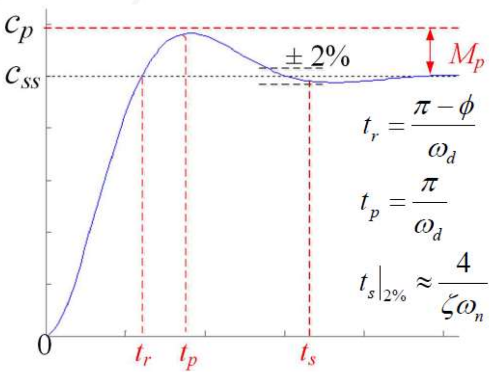
$\varsigma = 0$ (undamped)

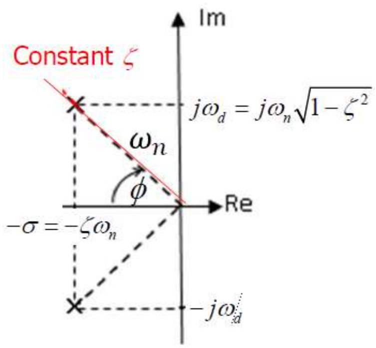
$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2+\omega_n^2}$$

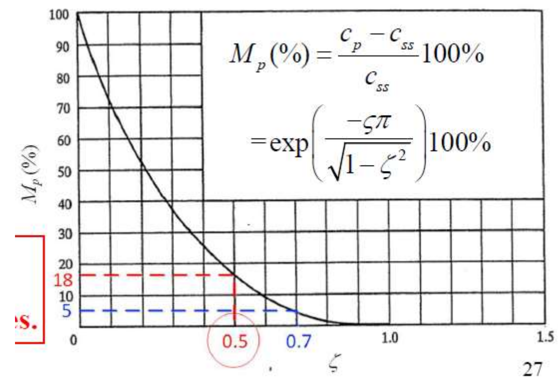


Over-damped Critically Damped Underdamped

Transfer Function - 3

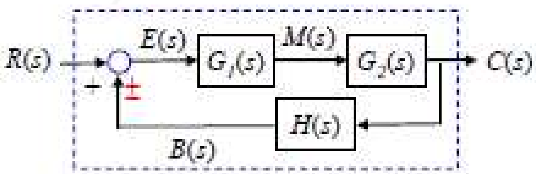




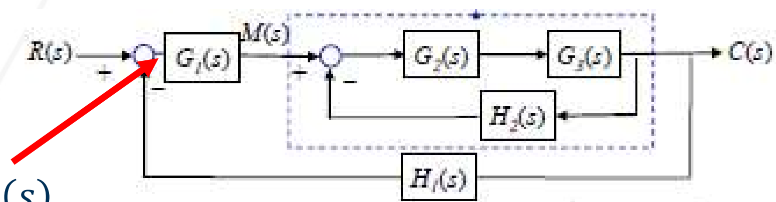


Block Diagram

1. Single Input / Single Loop

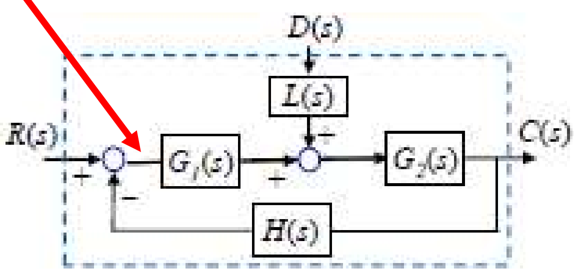


2. Single Input / Multiple Loop

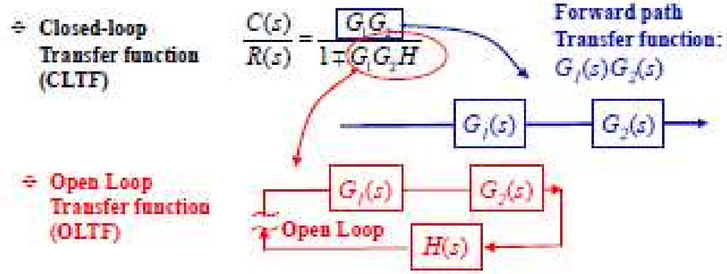


$E(s)$

3. Multiple Input / Single Loop



Transfer function of closed loop transfer function



Super Position

- $C(s)=\dfrac{G_1G_2}{1+HG_1G_2}R(s)+\dfrac{LG_2}{1+HG_1G_2}D(s)$

Same system characteristics
(independent of input and output)

- $E(s) = R(s) - C(s)$

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Stability Analysis

Routh stability criterion

- **Necessary and sufficient conditions:** The system is stable if and only if all the 1st elements in the 1st column of the Routh array are positive. $\boxed{R_{i1} > 0 \ (i = i, \cdots, n+1)}$

Characteristic equation: $a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = 0$

s^n	$R_{11} = a_n$	$R_{12} = a_{n-2}$	$R_{13} = a_{n-4}$	$\cdots$
s^{n-1}	$R_{21} = a_{n-1}$	$R_{22} = a_{n-3}$	$R_{23} = a_{n-5}$	$\cdots$
s^{n-2}	$R_{31} = \dfrac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}}$	$R_{32} = \dfrac{a_{n-1}a_{n-4} - a_n a_{n-5}}{a_{n-1}}$	$R_{33} = \dfrac{a_{n-1}a_{n-6} - a_n a_{n-7}}{a_{n-1}}$	$\cdots$
s^{n-3}	$R_{41} = \dfrac{R_{31}R_{22} - R_{21}R_{32}}{R_{31}}$	$R_{42} = \dfrac{R_{31}R_{23} - R_{21}R_{33}}{R_{31}}$	$R_{43} = \dfrac{R_{31}R_{24} - R_{21}R_{34}}{R_{31}}$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	
s^1	R_{n1}	0	0	
s^0	$R_{(n+1)1}$	0	0	

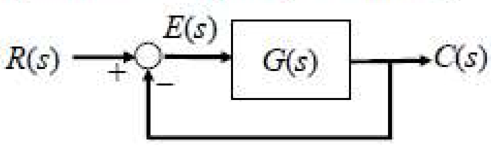
Case 1: An element in the 1st column is zero
- Replace 0 with ε
  - Cannot replace 0 in the first two rows

Case 2: The Routh Table has a row of 0 values.
- Take the derivative

Steady State Errors

Steady-state tracking error

- Specification (unity feedback)



$$G(s) = G_c(s)G_p(s)$$

$$E(s) = R(s) - C(s)$$

$$\frac{E(s)}{R(s)} = 1 - \frac{G(s)}{1+G(s)} = \frac{1}{1+G(s)}$$

If $e(\infty)$ exist,

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}$$

- System types

Step ③ Static Gain $K = \lim_{s\to 0} s^N G(s)$

$$G(s) = \frac{K}{s^N}\frac{a_k s^k + \cdots + a_1 s + 1}{b_l s^l + \cdots + b_1 s + 1}$$

Step ① Step ② Type number N (number of integrator).

$$s^N G(s) = K\left(\frac{a_k s^k + \cdots + a_1 s + 1}{b_l s^l + \cdots + b_1 s + 1}\right)$$

$$\lim_{s\to 0} s^N G(s) = K \lim_{s\to 0}\left(\frac{a_k s^k + \cdots + a_1 s + 1}{b_l s^l + \cdots + b_1 s + 1}\right) = K$$

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Steady-state tracking error

- Input types

$$r(t) = \begin{cases} 1(t) & m = 0,\ \text{unit-step} \\ t & m = 1,\ \text{unit-ramp} \\ \frac{1}{2}t^2 & m = 2,\ \text{unit-parabolic} \end{cases}$$

$$R(s) = \left[\frac{t^m}{m!}\right] = \frac{1}{m!}\mathcal{L}\left[t^m\right] = \frac{1}{s^{m+1}}$$

If $e(\infty)$ exist,

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0}\frac{sR(s)}{1+G(s)} = \lim_{s\to 0}\left[\frac{s}{1+G(s)}\frac{1}{s^{m+1}}\right] = \lim_{s\to 0}\frac{1}{s^m + s^m G(s)}$$

$$e_{ss} = \begin{cases} \dfrac{1}{1+\lim_{s\to 0} G(s)} & m = 0 \\ \dfrac{1}{\lim_{s\to 0} sG(s)} & m = 1 \\ \dfrac{1}{\lim_{s\to 0} s^2 G(s)} & m = 2 \end{cases}$$

Pos. error constant, $K_{pos} = K|_{m=0}$

Vel. error constant, $K_{vel} = K|_{m=1}$

Acc. error constant, $K_{acc} = K|_{m=2}$

Input type m

Static Gain $K = \lim_{s\to 0} s^m G(s)$

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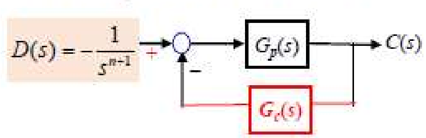
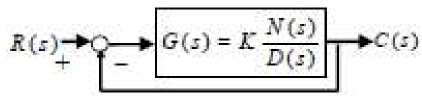
Steady-state error due to disturbance (R=0)



$$\frac{E(s)}{D(s)} = \frac{-C(s)}{D(s)} = \frac{-G_p}{1+G_cG_p}$$

Root Locus

Root locus (RL) is a locus of the roots of the characteristic equation plotted in the s-plane as a function of K.



OL pole (×)
OL zero (○)
CL pole (□)

CLCE: $D(s) + KN(s) = 0$ where $0 \le K < \infty$.

Notes 1) K=0, $D(s) = 0$ The roots are also the POLEs of the OLTF.

2) $K \to \infty$, $\frac{D(s)}{K} + N(s) = 0 \to N(s) = 0$ The roots approaches to the ZEROs nof the OLTF.

3) CLCE: $\frac{KN(s)}{D(s)} = -1 + j0$

Points on RL satisfy two conditions:

Mag. $K\left|\frac{N(s)}{D(s)}\right| = 1 \Rightarrow K = \frac{|D(s)|}{|N(s)|}$

Phase $\angle\left[\frac{N(s)}{D(s)}\right] = \begin{cases} \pm 180(2l+1) & K \ge 0 \\ \pm 180(2l) & K < 0 \end{cases}$ where $l = 0, 1, 2, \cdots$

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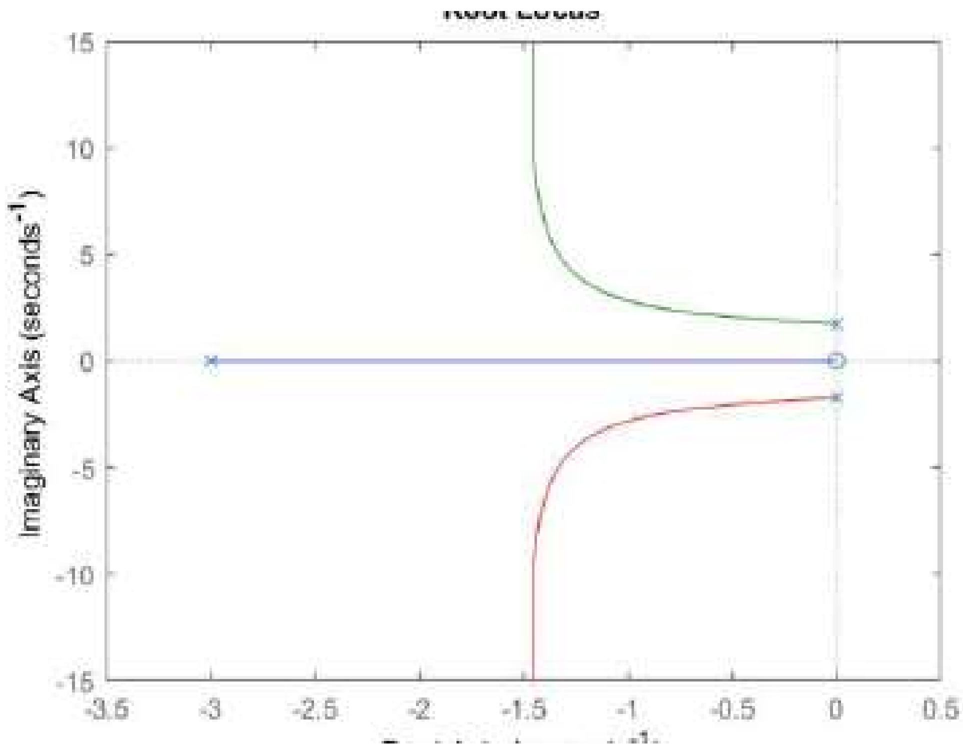

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Root Locus Sketching

How to sketch root-locus plot ($0 < K < \infty$)

❑ Express the CLCE in standard form before you start.

A. Quick sketches:

1) Locate all n OL poles and m zeros on s-plane.

   # of RL branches $= n$

   Rule 2: Left of ODD number of OL poles/zeros.

2) Determine the root loci on the real axis and the asymptotes; Rule 2 and Rule 3 (n–m).

   Rule 3

   $$\sigma_c = \frac{\sum_{i=1}^{n}(-p_i) - \sum_{j=1}^{m}(-z_j)}{n-m}$$

   $$\beta = \pm\frac{180^\circ(2l+1)}{n-m}$$
   where $l = 0, 1, 2, \ldots$

3) Complete all branches **without detailed calculations.**

B. Calculations on specific points on RL:

4) Find the breakaway and break-in points; Rule 4 (select the solutions for $K>0$ or $K<0$.

   $$\left.\frac{dK}{d\sigma}\right|_{s=\sigma_b} = \left.\frac{dG^{-1}}{d\sigma}\right|_{s=\sigma_b} = 0$$ or $$\sum_{i=1}^{n}\frac{1}{\sigma_b - p_i} = \sum_{j=1}^{m}\frac{1}{\sigma_b - z_j}$$

5) Determine the angle of departure (arrival) of the roolocus from a complex pole p (zero, z); Rule 5.

   $$\theta_D\big|_p = 180^\circ + \underbrace{\angle G'}_{\angle G \text{ ignoring } p}$$

   $$\theta_A\big|_z = 180^\circ - \underbrace{\angle G''}_{\angle G \text{ ignoring } z}$$

6) Use Routh criterion to determine the point at which the RL crosses the $j\omega$ axis.

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